

Restock a Condiment Caddy SOP (1x Caddy)

One episode restocks one condiment caddy. Work the four phases in this fixed order: **refill, wipe and cap the three bottles, load the caddy by slot, fold the napkin stack into the napkin bay, top up the stirrers**. The three condiments, the slot labels, and the order of the phases never vary.

This is a single-arm task. Only the **right arm** works. The **left arm stays at home with its gripper open for the whole episode** and touches nothing. Because there is no second gripper to hold or steady anything, the fixtures do that job: a bottle is worked on only while it stands in the **bottle well**, the caddy stands on its mark and is never gripped, and the napkin stack is folded on a board that holds its back half down. Nothing is ever held in the air while something else is done to it.

Every bottle is poured, wiped, and capped standing in its well hole. A bottle is lifted only once, when it goes into the caddy. The bottles start with their labels facing the front edge and keep facing that way to the end, so the right gripper never has to turn one.

The table is set up in one of three ways. Only the **bottle well**, with the three bottles standing in it, moves; the jug shelf, the cloth rest, the caddy, the fold board, and the stirrer tub are in the same place in all three. The cap tray stands beside the well and changes sides in Config R.

- **Config L:** the bottle well is at the center-left, between the jug shelf and the fold board. The cap tray is right of the well.
- **Config M:** the bottle well is at the center of the table. The cap tray is right of the well.
- **Config R:** the bottle well is at the center-right, between the stirrer tub and the cloth rest. The cap tray is left of the well.

Where a step depends on the setup it says so on an **IF** line — look at the table and follow the line that matches.

What stays constant across all sessions:

- **Start position:** the bottle well stands at the center-left (**Config L**), the center (**Config M**), or the center-right (**Config R**). One config per episode, chosen before recording and never changed mid-episode.
- **Single arm:** the **right gripper** does all the work in all three configs, and the **left arm stays parked** even when the well is on the left. The config changes only which way the right gripper reaches for the well and the cap tray. Nothing is handed over.
- **Supply side:** the cap tray is on the right of the well in Config L and M, and on the left of the well in Config R.

- **Fixed roles:** ketchup, then mustard, then hot sauce; refill, wipe and cap, load the caddy, fold the napkins, top up the stirrers; every bottle is worked standing in its well hole and lifted only once, into its matching slot.

Setup

Complete both checklists before recording starts.

Cell configuration

- **Environment camera:** 900 mm.
- **cell_type:** single_arm

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the jug shelf across the back left, the stirrer tub at the back right, the bottle well in the start zone for this episode's config, the cap tray beside it on its supply side, the cloth rest at the front right, the caddy on its mark at the front center, the fold board at the front left, and the front table edge.
3. The three bottles are visible from the side, so the sauce level and the fill line on each one can be seen.
4. The caddy is visible from above, so all three slot labels, the napkin bay, and the inside of the stirrer cup can be seen.
5. The fold board is visible from above, so the crease line, the back stop, and the napkin stack can be seen.
6. Both arms are at home with grippers open.
7. The table is clean, dry, and bare apart from the jug shelf and jugs, the bottle well and bottles, the cap tray and caps, the cloth rest and cloth, the caddy, the fold board and napkin stack, and the stirrer tub.
8. The right arm reaches all three jugs, all three well holes (Config L, M, and R), the cap tray on either side of the well, the cloth rest, all three caddy slots, the napkin bay, the stirrer cup, the fold board, and the stirrer tub without stretching.
9. The left arm is at home, gripper open, and stays there.

Materials checklist

1. **Three condiment bottles**, clear, straight-sided, each with a **fill line** marked around the outside and a colored label: **ketchup** (red), **mustard** (yellow), **hot sauce** (orange).
2. All three bottles stand **uncapped** in the bottle well at the start, one per hole, left to right: ketchup, mustard, hot sauce. Each label faces the front edge.
3. Each bottle starts holding a little sauce, with its level clearly **below** the fill line.

4. The **bottle well** is one heavy block with three upright holes in a row. Each hole is labelled with the condiment name and its color. The block is heavy enough that a bottle can be poured into, wiped, and capped without the block sliding. It stands in the start zone for the config:
 - **Config L:** center-left, between the jug shelf and the fold board.
 - **Config M:** center of the table.
 - **Config R:** center-right, between the stirrer tub and the cloth rest.
5. **Three refill jugs** stand upright on the **jug shelf** across the back left, left to right: ketchup, mustard, hot sauce. Each jug has a handle, a pouring spout, and a colored band matching its bottle label. Each jug holds enough to fill its bottle to the fill line with plenty to spare, and is light enough for one gripper.
6. Each jug stands on its own **jug spot**, a taped outline on the shelf, handle toward the front edge.
7. **Three flip caps** lie dome up on the **cap tray**, beside the bottle well on its supply side (right of the well in Config L and M, left of it in Config R), in a row left to right: red, yellow, orange. A cap is pressed straight down onto a bottle; it does not screw on.
8. One folded **damp cloth** per episode lies flat on the **cloth rest** at the front right. It is wrung out so it does not drip. A cloth is used for one episode and thrown away.
9. One **caddy** stands on the **caddy mark**, a taped outline at the front center. Left to right the caddy holds: the **napkin bay**, then **slot 1 (ketchup)**, **slot 2 (mustard)**, **slot 3 (hot sauce)**, then the **stirrer socket**. Each slot carries a label with the condiment name and its color. All three slots start empty and the napkin bay starts empty.
10. The caddy base is heavy and non-slip, so a bottle can be lowered into a slot without the caddy sliding.
11. The **stirrer cup** stands in the stirrer socket at the right end of the caddy. It starts holding a few stirrers, standing upright, with bare cup floor showing around them. The cup stays in its socket for the whole episode.
12. The **fold board** lies flat at the front left. It carries a **back stop** along its back edge, a **grip strip** of rubber across its back half, and a **crease line** marked across its middle.
13. One **napkin stack** of **six paper napkins** lies flat on the fold board, edges stacked square, back edge against the back stop, back half resting on the grip strip, and the stack covering the crease line.
14. The **stirrer tub** sits at the back right, open top, holding stirrers standing upright and packed together, and nothing else. It holds far more stirrers than one episode needs.

Workspace layout

Everything below is a fixed area of the table, judged by eye against the table edges, the taped outlines, and the fixtures standing on it.

- **Back left — jug shelf:** three jug spots in a row, left to right: ketchup, mustard, hot sauce.
- **Back right — stirrer tub:** open top, stirrers standing upright inside.
- **Start zone — bottle well:** center-left (Config L), center (Config M), or center-right (Config R). Three labelled holes in a row, left to right: ketchup, mustard, hot sauce. Every pour, wipe, and cap happens here.
- **Supply side — cap tray:** right of the well in Config L and M, left of the well in Config R. Three flip caps lying dome up, left to right: red, yellow, orange.
- **Front right — cloth rest:** the damp cloth, folded flat.
- **Front center — caddy mark:** one taped outline. The caddy stands here and does not move.
- **Front left — fold board:** the napkin stack, its back edge against the back stop.

Arm assignment

- **Right gripper — all of the work:** pours from each jug, wipes each bottle, presses each cap on, lifts each bottle into its caddy slot, folds the napkin stack and loads the napkin bay, and moves stirrers from the tub into the stirrer cup, in all three configs. In Config L it reaches left for the well, in Config R it reaches left of the well for the cap tray.
- **Left arm — parked:** stays at home with its gripper open from the start of recording to the end, in all three configs. It holds nothing, steadies nothing, and reaches for nothing.

Vocabulary

- **Bottle well:** the heavy block with three labelled holes. A bottle stands in a hole while it is poured into, wiped, and capped.
- **Start zone:** where the bottle well stands at the start of the episode — center-left (**Config L**), center (**Config M**), or center-right (**Config R**). One per episode, chosen before recording and never changed mid-episode.
- **Supply side:** the side of the well the cap tray stands on — the right in Config L and M, the left in Config R.
- **Fill line:** the line marked around the outside of a bottle. It is the target level for the refill.
- **Level at the fill line:** seen from the side, the top of the sauce sits on the fill line. Not visibly above it and not visibly below it.
- **Jug spot:** a taped outline on the jug shelf for one named jug. The jug ends the episode standing inside its own outline, upright, handle toward the front edge.
- **Matching jug:** the jug whose colored band is the same color as the bottle's label. Red to ketchup, yellow to mustard, orange to hot sauce. No other test is applied.
- **Matching cap:** the cap that is the same color as the bottle's label.
- **Seated cap:** the cap sits flat on the bottle, level, with no gap showing between the cap and the bottle collar anywhere around it.

- **Clean outside:** seen from the side, no run, streak, or wet patch shows anywhere on the outside of the bottle from the collar down to its base.
- **Wipe stroke:** one pass of the cloth, pressed flat against the bottle, drawn straight down from just under the collar to the base without lifting off part way.
- **Damp cloth:** the cloth is wet enough to pick up a run of sauce but wrung out so no drop falls from it while it is carried.
- **Slot label:** the name and color printed at each caddy slot. A bottle belongs in the slot whose label matches its own.
- **Label facing the front edge:** the bottle's printed label looks square to the front table edge, readable from the front.
- **Caddy mark:** the taped outline at the front center. The caddy stands inside it all episode. A correctly placed caddy shows a thin band of tape all the way around its base.
- **Fold board:** the flat board at the front left where the napkin stack is folded.
- **Back stop:** the raised lip along the back edge of the fold board. Napkin edges are squared against it.
- **Grip strip:** the rubber strip across the back half of the fold board. It holds the back half of the napkin stack down while the front half is lifted and folded over.
- **Crease line:** the line marked across the middle of the fold board. The fold is made on this line.
- **Layers together:** all six napkins move as one. No napkin is left behind on the board and no napkin slides out of line with the rest.
- **Folded edge:** the closed edge of the folded stack, the one made by the fold. The opposite edge is open and shows the layers.
- **Cup full:** looking down into the stirrer cup from above, stirrers fill it side to side and no bare cup floor shows around them.
- **Stirrer group:** the stirrers taken in one closing of the right gripper. The number in a group is not fixed and is never counted.
- **Standing upright:** every stirrer in the cup points up, none lies across the others, and none hangs over the cup rim.
- **Stable placement:** the item stays still for 2 seconds after release and does not rock, roll, tip, or slide.
- **Parked arm:** the left arm, at home with its gripper open. It does not move during the episode.

Steps

Steps 1.1, 1.3, 2, and 4 depend on where the bottle well stands: the jug is carried to it, the cap is fetched from its supply side, each bottle is carried from it to the caddy, and in Config R the stirrers are carried past it. Every other line is the same in all three configs.

Step 1: Refill, wipe, and cap one bottle

Goal: one bottle stands in its well hole, filled to its fill line, clean on the outside, with its matching cap seated.

Work the bottles in this fixed order: **ketchup, then mustard, then hot sauce**. Run all three parts of this step on one bottle before starting the next bottle. Leave the bottle standing in its well hole for the whole step. The **right gripper** does every action.

1.1 Pour the refill

- With the **right gripper**, grasp the matching jug by its handle.
- Lift it straight up off the shelf and keep it upright while carrying it.
- **IF the well is at the center-left (Config L):** carry the jug straight forward off the shelf to the bottle, clear of the fold board.
- **IF the well is at the center (Config M):** carry the jug forward and to the right to the bottle.
- **IF the well is at the center-right (Config R):** carry the jug across to the right to the bottle, clear of the cap tray on the near side of the well.
- Then, in all three: bring it over the bottle and lower it until the spout is just above the bottle mouth.
- Tip the jug slowly until the sauce runs into the bottle. Keep the spout over the mouth for the whole pour.
- Watch the level rise inside the bottle. Stop pouring when the level reaches the fill line.
- Tip the jug back upright and hold it over the bottle mouth for 2 seconds, so the last drips fall inside the bottle.
- Lift the jug clear, carry it back to its own jug spot, stand it upright with its handle toward the front edge, and release it when it is stable.

Check: seen from the side, the level sits on the fill line. If the level is below the line, pick up the same jug with the **right gripper** and pour again. If the level went above the line, do not pour more and do not try to take any out; go on to 1.2.

Expected state: the bottle stands in its hole, filled to the fill line, still uncapped. The jug is back on its spot.

1.2 Wipe the bottle

- With the **right gripper**, pick the cloth up off its rest at the middle of the cloth.
- Press it flat against the bottle just under the collar and draw it straight down to the base. That is one wipe stroke.
- Wipe four strokes on the bottle, in this order: the front, the right side, the back, the left side.
- Do not lift the cloth part way through a stroke.
- Carry the cloth back to its rest with the **right gripper**, lay it flat, and release.

Check: the outside of the bottle is clean from the collar down to its base. If a run or wet patch is still showing, pick the cloth up again with the **right gripper** and wipe that side once more. If a drip landed on the top of the well, wipe it off with the same cloth before putting the cloth back.

Expected state: the bottle stands filled and clean in its hole. The cloth is flat on its rest.

1.3 Press the cap on

- **IF Config L or M:** the cap tray is on the right of the well; reach right for the cap. **IF Config R:** the cap tray is on the left of the well; reach left for the cap.
- With the **right gripper**, grasp the matching cap around its outside on the cap tray.
- Lift it clear of the tray and keep it dome up.
- Bring it over the bottle and lower it straight down until it touches the bottle mouth.
- Press straight down until the cap sits flat against the bottle collar. Do not turn or twist the cap.
- Release the cap and lift the gripper clear.

Check: the cap sits level, with no gap showing between the cap and the bottle collar anywhere around it. If a gap shows, close the **right gripper** on the cap again and press straight down once more. Do not twist it.

Expected state: the bottle stands in its hole, filled, clean, and capped.

Repeat Step 1 for the mustard bottle, then for the hot sauce bottle. Only when all three bottles are filled, wiped, and capped does Step 2 start.

Step 2: Load the caddy by slot

Goal: all three bottles stand upright in their matching caddy slots, labels facing the front edge.

Load them in this fixed order: **ketchup, then mustard, then hot sauce**. Use the **right gripper** for all three. Do not grip, lift, or push the caddy at any point.

- With the **right gripper**, grasp the bottle around its middle. Do not grasp it by the cap and do not grasp it by the neck.
- Lift it straight up out of the well hole and keep it upright.
- **IF Config L:** the well is left of the caddy; bring the bottle in behind the caddy, not across the napkin bay. **IF Config M:** bring it straight forward to the caddy. **IF Config R:** the well is right of the caddy; bring the bottle in behind the caddy, not across the stirrer cup.
- Carry it to the caddy slot whose label matches the bottle's label. Do not pass it over the napkin bay or over the stirrer cup.
- Keep the label facing the front edge for the whole carry.
- Lower the bottle into the slot until it stands on the slot floor, and release it when it is stable.

Check: each bottle stands upright in the slot whose label matches it, with its label facing the front edge and its cap on. If a bottle is in the wrong slot, lift it out with the **right gripper**, grasp it at its middle, and set it into the right one. If its label is turned away, lift it clear of the slot and lower it back in with the label facing front.

Expected state: all three well holes are empty. The caddy holds ketchup, mustard, and hot sauce in slots 1, 2, and 3. The napkin bay is still empty. The caddy is still inside its mark.

Step 3: Fold the napkin stack and load the napkin bay

Goal: the six napkins are folded together as one stack and the folded stack sits in the napkin bay, folded edge toward the front edge.

3.1 Fold the stack

- With the **right gripper**, pinch the napkin stack at the middle of its front edge, closing on all six layers.
- Lift the front edge up and carry it back over the crease line, keeping it low over the board.
- Lay it down on the back half of the stack so the front edge comes to rest against the back stop.
- Release and lift the gripper clear.

Check: all six layers folded together, the lifted edge is against the back stop, and the side edges of the fold line up with the side edges underneath. If a napkin was left behind on the board, or a layer sticks out past the others, unfold the stack with the **right gripper**, press its edges square against the back stop, and fold it again.

3.2 Press the crease

- With the **right gripper** closed, press down on the crease at its left end.
- Draw the closed gripper along the crease from left to right in one pass, staying on the crease.
- Lift the gripper clear at the right end.

Check: the crease lies flat along the crease line, with no layer standing up off the board. If a layer stands up, press along the crease once more.

3.3 Load the napkin bay

- With the **right gripper**, close on the folded stack at the middle of its folded edge, taking the full thickness.
- Lift it straight up off the grip strip, keeping it flat.
- Carry it to the napkin bay at the left end of the caddy. Do not pass it over the bottle slots.
- Lower it into the bay with the folded edge toward the front edge, and release when it is stable.

Check: the folded stack sits inside the bay, flat, with its folded edge toward the front edge and no napkin hanging over the bay wall. If a napkin slid out during the carry, pick it up with the **right gripper** and put it in the bay on top of the stack.

Expected state: the fold board is bare. The napkin bay holds the folded stack. The three bottles are untouched in their slots.

Step 4: Top up the stirrers

Goal: the stirrer cup is full of stirrers standing upright, and the cup has not left its socket.

- Leave the stirrer cup in its socket. Do not grip it, lift it, or push it.
- With the **right gripper**, close on the stirrers in the tub at their **upper ends**, taking one stirrer group.
- Lift the group straight up out of the tub, keeping the stirrers upright.
- **IF Config R:** the empty well stands between the tub and the cup; carry the group down the right side of the well, not over its holes.
- Carry it over the stirrer cup and lower it until the lower ends of the stirrers are inside the cup.
- Release the group and lift the gripper clear.
- Look down into the cup. If bare cup floor still shows around the stirrers, take another group from the tub the same way.
- Keep taking groups until the cup is full. The number of groups is not fixed and is never counted.
- If a stirrer falls on the table or into the caddy, pick it up with the **right gripper** and stand it in the cup.

Check: looking down into the cup, stirrers fill it side to side with no bare cup floor showing, and every stirrer stands upright with none lying across the others or hanging over the rim. If a stirrer lies across the top, pick it up with the **right gripper** and stand it in the cup.

Expected state: the cup stands full and upright in its socket. The tub still holds stirrers. No stirrer is loose on the table or in the caddy.

Step 5: End the episode

Goal: recording ends with the caddy restocked.

1. Confirm the end state:

- all three bottles stand upright in their matching slots, capped, labels facing the front edge, each filled to its fill line and clean on the outside;
- the folded napkin stack sits flat in the napkin bay, folded edge toward the front edge;
- the stirrer cup stands full and upright in its socket, stirrers upright;
- all three well holes are empty and the fold board is bare;

- all three jugs stand upright on their own spots and the cloth lies flat on its rest;
 - the caddy is still inside its mark and the bottle well has not moved;
 - no sauce run, loose napkin, or loose stirrer is left on the table.
2. Return the right arm home with its gripper open. The left arm is already home. Homing is the last thing the arm does.
 3. Stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the three bottles out of the caddy slots. Take their caps off and lay the caps dome up on the cap tray, left to right: red, yellow, orange.
2. Stand the bottle well in the start zone for the next episode's config — center-left (Config L), center (Config M), or center-right (Config R) — with the cap tray on its supply side. Pour each bottle back down to a level clearly below its fill line, and stand it in its own well hole with its label facing the front edge.
3. Wipe each bottle dry on the outside and wipe the top of the bottle well dry.
4. Top the three jugs back up, and stand each one upright on its own jug spot with its handle toward the front edge.
5. Take the folded napkin stack out of the napkin bay, unfold it, and lay six napkins flat on the fold board, edges square, back edge against the back stop and back half on the grip strip.
6. Take stirrers out of the stirrer cup and stand them back in the stirrer tub until only a few are left in the cup, with bare cup floor showing around them. Leave the cup in its socket.
7. Throw the used cloth away. Lay a fresh damp cloth, wrung out so it does not drip, flat on the cloth rest.
8. Wipe the table dry and clear away any loose napkin or stirrer.
9. Stand the caddy inside its mark, with a thin band of tape showing all the way around its base.
10. Check the caddy mark and the jug spots. Re-tape any outline that is lifting, torn, or unreadable.
11. Inspect the bottles, caps, jugs, well, caddy, cup, tub, and fold board for damage. Replace damaged items.
12. Run both Setup checklists again.

SOP violations

Things that break this SOP and that reviewers look for in the side-by-side review tool.

How to record a violation in review

For every violation seen in a recorded episode, record:

- the **start timestamp** in the video;
- the **violation name** from the list below; and
- the **SOP rule broken**, including the step number.

The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. An episode may contain several violations; tag each one separately. Retain the episode in training data with its violation tags. Do not delete a recorded episode solely because it contains a violation.

Violations

Note on the start position: the violations below were written for Config M (the bottle well at the center of the table, cap tray on its right). The pickup and arm-role cues will be rewritten later to cover all three start positions; they are left as they are for now. Until then, anything that does not match the episode's config goes under **Config misaligned**.

Violation: Config misaligned

- **Visible cue:** what the operator does does not match the config on the table — the bottle well is not in the start zone for the config; the cap tray is not on the well's supply side; the right gripper reaches to the wrong side of the well for a cap; or the wrong IF line is followed.
- **SOP rule broken:** the start position and the supply side (the well stands at the center-left, center, or center-right for the episode's config, the cap tray on its right in Config L and M and on its left in Config R; the IF line followed is the one for the config on the table).
- **Coaching note:** look where the well and the cap tray are before the first reach, then follow that config's IF lines through Steps 1.1, 1.3, 2, and 4.

Violation: Wrong jug used

- **Visible cue:** the right gripper pours into a bottle from a jug whose colored band does not match that bottle's label, for example the yellow-banded jug over the ketchup bottle.
- **SOP rule broken:** Step 1.1, each bottle is filled from the jug whose band matches its label.
- **Coaching note:** check the band color against the label color before lifting the jug.

Violation: Fixed left-to-right order not followed

- **Visible cue:** mustard or hot sauce is refilled before ketchup, hot sauce is refilled before mustard, or the bottles go into the caddy slots in any order other than ketchup, mustard, hot sauce.

- **SOP rule broken:** Steps 1 and 2, bottles are worked and loaded in the fixed order ketchup, then mustard, then hot sauce.
- **Coaching note:** ketchup, mustard, hot sauce, every time, whichever bottle is nearest.

Violation: Wrong phase order

- **Visible cue:** a cap goes on before the bottle is wiped, a bottle goes into a caddy slot before all three bottles are capped, the napkins are folded before the caddy is loaded, or the stirrers are topped up before the napkin stack is in its bay.
- **SOP rule broken:** Steps 1 to 4, work the four phases in the fixed order: refill, wipe and cap all three bottles, load the caddy, load the napkins, top up the stirrers.
- **Coaching note:** finish each phase and its check before starting the next one.

Violation: Pour done wrong

- **Visible cue:** the right gripper tips the jug before the spout is over the bottle mouth, pours from high above the mouth, lets the spout drift off the mouth mid-pour, carries the jug tilted, or leaves spilled sauce on the well or the table.
- **SOP rule broken:** Step 1.1 (carry the jug upright, lower the spout to just above the bottle mouth, and keep the spout over the mouth for the whole pour).
- **Coaching note:** get the spout over the mouth first, then tip, and keep it there until the jug is upright again.

Violation: Fill level wrong

- **Visible cue:** at the end of Step 1.1 the sauce level sits visibly above the fill line, or sits visibly below it and no second pour follows.
- **SOP rule broken:** Step 1.1 (stop pouring when the level reaches the fill line, and pour again if it is below).
- **Coaching note:** watch the level from the side during the pour, not the jug.

Violation: Bottle held or lifted while being worked on

- **Visible cue:** the right gripper lifts a bottle out of its well hole, or holds or steadies it, during the pour, the wipe, or the cap press.
- **SOP rule broken:** Steps 1.1, 1.2 and 1.3, the bottle stays standing in its well hole for the whole of Step 1. It is lifted only in Step 2.
- **Coaching note:** the well holds the bottle. The gripper's only job in Step 1 is the jug, the cloth, and the cap.

Violation: Jug not returned to its spot

- **Visible cue:** a jug is left standing on the table, set down off its outline, set down on the wrong outline, laid on its side, or left with its handle turned away from the front edge.

- **SOP rule broken:** Step 1.1 (carry the jug back to its own jug spot, upright, handle toward the front edge).
- **Coaching note:** the pour is not finished until the jug is back inside its outline.

Violation: Bottle not wiped, or wiped wrong

- **Visible cue:** a cap goes on with a run, streak, or wet patch still showing on the bottle; or the cloth is dabbed instead of drawn down, fewer than four strokes are made, a stroke misses a side of the bottle, or the cloth lifts off part way down a stroke.
- **SOP rule broken:** Step 1.2 (four wipe strokes, front, right side, back, left side, each drawn straight down from under the collar to the base without lifting).
- **Coaching note:** press flat and pull straight down, four sides, then look before reaching for the cap.

Violation: Cloth misused or left off its rest

- **Visible cue:** the cloth is run over the caddy, the napkin stack, the cap tray, or the bare table; the cloth is dropped on the table; or the cloth is not laid back flat on its rest before the cap is picked up.
- **SOP rule broken:** Step 1.2, the cloth wipes the bottle and the top of the well only, and goes back flat on its rest at the end of the wipe.
- **Coaching note:** the cloth touches the bottle and the well, nothing else, and it ends every wipe on its rest.

Violation: Wrong cap on a bottle

- **Visible cue:** a cap is pressed onto a bottle whose label is a different color from the cap, for example the orange cap on the mustard bottle.
- **SOP rule broken:** Step 1.3, each bottle takes the cap that matches its label color.
- **Coaching note:** match the cap color to the label color on the tray before lifting it.

Violation: Cap not seated, or twisted on

- **Visible cue:** the cap is released sitting tilted, or with a gap showing between the cap and the bottle collar; the right gripper turns or twists the cap instead of pressing it straight down; or the cap falls off the bottle later in the episode.
- **SOP rule broken:** Step 1.3 (lower the cap straight down and press until it sits flat against the collar with no gap, without turning it).
- **Coaching note:** straight down and press. These caps click on, they do not screw on.

Violation: Bottle put in the wrong slot

- **Visible cue:** a bottle is released in a slot whose label does not match its own label, or is released in the napkin bay or the stirrer socket.

- **SOP rule broken:** Step 2, each bottle goes in the slot whose label matches its label.
- **Coaching note:** read the slot label on the way down, not after the release.

Violation: Bottle carried or set down wrong

- **Visible cue:** a bottle is grasped by its cap or its neck, dragged or slid across the table instead of lifted clear, carried tilted, carried over the napkin bay or the stirrer cup, released before it is stable, released leaning, or released with its label turned away from the front edge.
- **SOP rule broken:** Step 2 (grasp the bottle at its middle, lift it straight up, keep it upright and label-forward, and release only when it is stable).
- **Coaching note:** middle grip, straight up, label to the front, and let go only once it stands on its own.

Violation: Napkin fold wrong

- **Visible cue:** the front edge is folded short of the back stop or past it, a napkin is left behind on the board, layers fan out so the side edges do not line up, the stack is dragged back instead of lifted over, or the fold is made anywhere but on the crease line.
- **SOP rule broken:** Step 3.1 (pinch the middle of the front edge through all six layers, carry it back over the crease line, and lay it against the back stop).
- **Coaching note:** pinch the middle so all six come, and aim the edge at the back stop.

Violation: Crease not pressed

- **Visible cue:** the right gripper goes straight from the fold to lifting the stack with no pass along the crease, presses only one spot, or the pass leaves the crease part way along.
- **SOP rule broken:** Step 3.2 (press down at the left end of the crease and draw along it to the right end in one pass).
- **Coaching note:** one steady pass, left to right, staying on the crease.

Violation: Napkin stack loaded wrong

- **Visible cue:** the folded stack is released with its open edge toward the front edge, is dropped into the bay from above, sits on the bay wall or hangs over it, is set on a bottle slot instead of the bay, or a napkin that slid out during the carry is left on the table.
- **SOP rule broken:** Step 3.3 (lift the stack by the middle of its folded edge, lower it into the bay with the folded edge toward the front edge, and pick up any napkin that slides out).
- **Coaching note:** folded edge to the front, lower it in, and look for strays before moving on.

Violation: Stirrer cup lifted or knocked

- **Visible cue:** the right gripper grips or lifts the stirrer cup out of its socket, or the cup is pushed, tipped, or knocked over while stirrers are being loaded.

- **SOP rule broken:** Step 4, the cup stays in its socket and is never gripped, lifted, or pushed.
- **Coaching note:** load into the cup where it stands. Come down over the middle of the cup, not over its rim.

Violation: Stirrer cup not filled

- **Visible cue:** the episode moves on to Step 5 while bare cup floor still shows around the stirrers when looking down into the cup.
- **SOP rule broken:** Step 4 (keep taking groups from the tub until no bare cup floor shows).
- **Coaching note:** look straight down into the cup after every group, and take another if floor still shows.

Violation: Stirrers loaded wrong

- **Visible cue:** the right gripper closes on the stirrers at their lower ends, releases a group above the cup so the stirrers drop in from height, carries a group tilted or lying flat, or leaves stirrers lying across the top of the cup or hanging over its rim.
- **SOP rule broken:** Step 4 (take each group at the upper ends, keep it upright, lower it until the lower ends are inside the cup, then release, and stand any stirrer that lies across the top).
- **Coaching note:** grip high, keep them upright, and put the ends in the cup before opening the gripper.

Violation: Caddy or bottle well moved

- **Visible cue:** the caddy shifts off its taped mark, or is gripped, lifted, or pushed; or the bottle well slides out of place while a bottle is poured into, wiped, or lifted out.
- **SOP rule broken:** Steps 1 to 4, the caddy stays inside its mark and the bottle well stays where it is. Neither is ever gripped or pushed.
- **Coaching note:** come down onto a slot or a hole from straight above, and do not lean the gripper against a fixture.

Violation: Parked arm moved

- **Visible cue:** the left arm leaves home, its gripper closes, or it reaches toward, holds, or steadies anything at any point in the episode.
- **SOP rule broken:** Steps 1 to 5, this is a single-arm task. The left arm stays at home with its gripper open from the start of recording to the end.
- **Coaching note:** the fixtures do the holding. If a step feels like it needs a second gripper, stop and report the fixture, do not use the left arm.

Violation: Item dropped or knocked over

- **Visible cue:** a jug, bottle, cap, napkin, or stirrer falls from the right gripper onto the table or the floor; a bottle is knocked over in the well or in a slot; a jug is knocked over on the shelf;

or the cap tray, the stirrer tub, or the fold board is knocked out of place.

- **SOP rule broken:** Steps 1 to 4 (lift each item clear, carry it on the path its step names, and release it only when it is stable).
- **Coaching note:** lift higher over the well and the caddy, and slow the carry near a fixture.

Violation: Finished item disturbed later

- **Visible cue:** a bottle already standing in its slot is bumped, turned, or knocked out; the napkin stack already in the bay is pushed out of place or pushed under a bottle; or a capped bottle loses its cap while a later step is worked.
- **SOP rule broken:** Steps 2 to 5, once an item is in its place in the caddy it stays there, untouched, for the rest of the episode.
- **Coaching note:** route the arm around the loaded slots and the bay once a phase is done.

Violation: Check skipped

- **Visible cue:** the right arm moves straight on to the next step with no look at the step's end state: no look at the fill line after a pour, no look at the bottle after the wipe, no look at the cap collar after the press, no look at the slot labels after loading, or no look into the stirrer cup after a group.
- **SOP rule broken:** Steps 1 to 4, each step's check is done before the next step starts.
- **Coaching note:** every step ends with a look, and the fix happens in that step, not later.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a bottle in the wrong slot, a bottle uncapped, unfilled, or dirty, the napkin bay empty, bare floor still showing in the stirrer cup, a jug off its spot, or sauce, a napkin, or a stirrer loose on the table; or the right arm is away from home, a gripper is closed, or an arm makes a correction after homing.
- **SOP rule broken:** Step 5 (confirm the end state, return the right arm home with its gripper open, then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arm does.

Non-violation failures

Failures that are not caused by how the task was run do not go in the violation set. Log them as system issues, discard the episode, and do not use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, collision caused by controller error, or motor error.
- **Defective item:** a cracked bottle or jug, a cap that will not seat on a sound bottle, a jug that will not pour, a caddy or well that slides on its own, a grip strip that no longer holds the

napkins, or a taped outline that lifts off the table during the episode. Replace before the next episode.

Annotation subtasks (from SOP)

1. Pour one bottle up to its fill line
2. Return one jug to its spot
3. Wipe one bottle on four sides
4. Return the cloth to its rest
5. Press one cap onto a bottle
6. Move one bottle from the well into its caddy slot
7. Fold the napkin stack over the crease line
8. Press the crease
9. Load the folded stack into the napkin bay
10. Move one stirrer group from the tub into the stirrer cup
11. Return the right arm home and end the episode